Hypothesis Testing (Part 3)

STAT 4530/6530
Statistical Inference for Data Scientists

Chi-Squared Tests for Multiple Proportions in a Contingency Table

Significance tests comparing two proportions are common in many application areas, such as comparing success rates of a new drug and placebo in a medical clinical trial.

- Sometimes the response variable has more than two categories or the comparison involves more than two treatments or groups, such as in comparing (placebo, low dose, high dose) of a drug on the response outcomes (better, same, worse).
- A natural null hypothesis is then that the probability distribution of the response variable is the same for each group, that is, the response variable is independent of the explanatory variable
- For a categorical response variable X_2 and categorical explanatory variable X_1, this is

$$H_0 : P(X_2 = j | X_1 = i) = P(X_2 = j) \text{ for all } i, j.$$

Chi-Squared Tests for Multiple Proportions in a Contingency Table

When both X_1 and X_2 are response variables, their joint distribution has cell probabilities p_{ij}. In that case, we can express the null hypothesis that X_1 and X_2 are independent as:

$$H_0 : p_{ij} = P(X_1 = i, X_2 = j) = P(X_1 = i)P(X_2 = j) \text{ for all } i, j.$$

Let y_{ij} denote cell counts in the contingency table that cross-classifies X_1 and X_2, with $\mu_{ij} = E[Y_{ij}]$ and total sample size n.

- The rows (with indices i) correspond to variable X_1 and the columns (with indices j) correspond to variable X_2.
- We denote the number of rows in the table by r and the number of columns by c.
- With $c > 2$, the multinomial distribution generalization of the binomial distribution applies to the c counts in row i having the conditional probabilities $P(X_2 = 1|X_1 = i), \ldots, P(X_2 = c|X_1 = i)$ for $i = 1, \ldots, r$.

Chi-Squared Tests for Multiple Proportions in a Contingency Table

Let $y_{i+} = \sum_j y_{ij}$ denote the row totals, and $y_{+j} = \sum_i y_{ij}$ denote the column totals of the sample marginal distributions.

The proportion y_{+j}/n of the observations fall in column category j, and under H_0, we expect each row to have this same proportion in that category.

- In other words, we expect the proportion y_{+j}/n of the y_{i+} observations to fall in the cell in column j, so we estimate the expected frequency μ_{ij} in that cell by

$$\hat{\mu}_{ij,0} = y_{i+} \frac{y_{+j}}{n}.$$

Chi-Squared Tests for Multiple Proportions in a Contingency Table

When both X_1 and X_2 are response variables, the maximum likelihood estimate of $P(X_1 = i)$ is y_{i+}/n, the maximum likelihood estimate of $P(X_2 = j)$ is y_{+j}/n, and the maximum likelihood estimate of $\mu_{ij} = np_{ij} = nP(X_1 = i)P(X_2 = j)$ is

$$\hat{\mu}_{ij,0} = n\hat{p}_{ij} = n\hat{P}(X_1 = i)\hat{P}(X_2 = j) = n\left(\frac{y_{i+}}{n}\right)\left(\frac{y_{+j}}{n}\right) = \frac{y_{i+}y_{+j}}{n}.$$

The test statistic for $H_0:$ independence summarizes differences between observed counts y_{ij} and estimated expected frequencies $\hat{\mu}_{ij,0}$ using

$$X^2 = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(y_{ij} - \hat{\mu}_{ij,0})^2}{\hat{\mu}_{ij,0}} = \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{[y_{ij} - (y_{i+}y_{+j})/n]^2}{(y_{i+}y_{+j})/n}.$$

For random counts Y_{ij}, the statistic can be expressed alternatively as a sum of squares of statistics having standard normal distributions, so its sampling distribution is the chi-squared distribution. The test statistic X^2 is known as the **Pearson chi-squared statistic**.

Chi-Squared Tests for Multiple Proportions in a Contingency Table

Greater differences $|y_{ij} - \hat{\mu}_{ij,0}|$ produce larger X^2 test statistic values and stronger evidence against H_0, so the P-value is the right-tail probability from the chi-squared distribution above the observed value.

- The chi-squared approximation improves as n increases, and it is usually adequate when $\hat{\mu}_{ij,0} \geq 5$.
- When a chi-squared test refers to multiple parameters, the degrees of freedom is the difference between the number of needed parameters under H_a and under H_0.

Chi-Squared Tests for Multiple Proportions in a Contingency Table

When both X_1 and X_2 are response variables and have a joint distribution, H_a has $rc - 1$ parameters (since the p_{ij} sum to 1 over the rc cells).

Under H_0, since $p_{ij} = P(X_1 = i)P(X_2 = j)$ for all i and j with $\sum_{i=1}^{r} P(X_1 = i) = 1$ and $P(X_2 = j) = 1$, H_0 has $(r-1) + (c-1)$ parameters.

Therefore, this chi-squared test has $df = (rc - 1) - (r - 1) - (c - 1) = (r - 1)(c - 1)$.

The same formula holds when only X_2 is a response variable and H_0 is homogeneity of the conditional distributions.

Example: Happiness and Marital Status

In their 2018 random sample of Americans, the General Social Survey queried subjects about their happiness and marital status.

We can use `R` to form the contingency table of counts y_{ij} cross-classifying happiness and marital status.

We then construct the sample conditional distribution of happiness for each marital status, find the estimates $\hat{\mu}_{ij,0}$ of the expected frequencies for testing $H_0:$ independence of happiness and marital status, and conduct the chi-squared test.

Example: Happiness and Marital Status

```
> Happy <- read.table("http://stat4ds.rwth-aachen.de/data/Happy.dat", header=TRUE)
# To construct contingency tables, define variables as factors
> Happiness <- factor(Happy$happiness); Marital <- factor(Happy$marital)
> levels(Happiness) <- c("Very happy", "Pretty happy", "Not too happy")
> levels(Marital) <- c("Married", "Divorced/Separated", "Never married")
> table(Marital, Happiness)              # forms contingency table
                    Happiness
Marital              Very happy Pretty happy Not too happy
  Married                   432          504            61
  Divorced/Separated         92          282           103
  Never married             124          409           135
> prop.table(table(Marital,Happiness), 1) # proportions within rows (each row total = 1)
                    Happiness             # conditional distributions on happiness
Marital              Very happy Pretty happy Not too happy
  Married            0.43329990   0.50551655    0.06118355
  Divorced/Separated 0.19287212   0.59119497    0.21593291
  Never married      0.18562874   0.61227545    0.20209581
> chisq.test(Marital, Happiness)$expected # expected frequencies under H0: independence
                    Happiness
Marital              Very happy Pretty happy Not too happy
  Married              301.6134     556.2162     139.17040
  Divorced/Separated   144.3025     266.1134      66.58403
  Never married        202.0840     372.6704      93.24556
> chisq.test(Marital, Happiness)          # chi-squared test of independence
        Pearson's Chi-squared test
X-squared = 197.41, df = 4, p-value < 2.2e-16
```

Example: Happiness and Marital Status

In their 2018 random sample of Americans, the General Social Survey queried subjects about their happiness and marital status.

We can use `R` to form the contingency table of counts y_{ij} cross-classifying happiness and marital status.

We then construct the sample conditional distribution of happiness for each marital status, find the estimates $\hat{\mu}_{ij,0}$ of the expected frequencies for testing $H_0:$ independence of happiness and marital status, and conduct the chi-squared test.

The Pearson chi-squared statistic is $X^2 = 197.4$, and because $r = c = 3$, we have $df = (r-1)(c-1) = 4$. A value of 197.4 has P-value essentially 0, which we can report as $P < 0.0001$. The evidence is extremely strong that happiness and marital status are not independent.

Significance Test Decisions and Errors

Significance tests report the P-value to summarize the evidence against H_0. When useful, a test can include a formal decision, rejecting H_0 if the P-value is sufficiently small.

- Before we observe the data, the P-value is a random variable.
- When H_0 is true, the P-value has approximately a uniform distribution between 0 and 1.
- When H_0 is false, the P-value is more likely to be near 0 than near 1. Researchers do not regard the evidence against H_0 as strong unless the P-value is very small (say, less than 0.05 or 0.01).
- We can base a decision about whether to reject H_0 on whether the P-value falls below a pre-specified cutoff point, called the α-level.
- The **α-level**, also called the **significance level**, is a number α between 0 and 1 such that we reject H_0 if the P-value is less than or equal to α. Common values for α are 0.05 and 0.01.

Significance Test Decisions and Errors

In practice, it is better to report the P-value than to indicate merely whether H_0 is rejected.

- The P-values of 0.049 and 0.001 both result in rejecting H_0 when $\alpha = 0.05$, but the second case provides much stronger evidence.
- P-values of 0.049 and 0.051 provide, in practical terms, the same amount of evidence about H_0.
- Most research articles report the P-value rather than a decision about H_0. From the P-value, readers can view the strength of evidence against H_0 and make their own decision, if they want to.

Significance Test Decisions and Errors

The example we previously covered about climate change tested $H_0 : p = 0.50$ about the population proportion p who thought climate change is a major threat to the stability of the country.

- The P-value of 0.21 was not small, so H_0 is plausible.
- In this case, the conclusion is sometimes reported as "do not reject H_0" or "we fail to reject H_0," since the data do not contradict H_0.
- It is however **not** appropriate to say "accept H_0."
- The population proportion has many plausible values besides the number in H_0. For instance the 95% confidence interval for p is (0.49, 0.55).
- Even though insufficient evidence exists to conclude that $p \neq 0.50$, it is improper to conclude that $p = 0.50$.

Significance Test Decisions and Errors

Null hypotheses such as $H_0 : p = 0.50$, $H_0 : \mu = 0$, $H_0 : p_1 - p_2 = 0$ and $H_0 : \mu_1 - \mu_2 = 0$ contain a single value for the parameter.

- When the P-value is larger than the α-level, saying "do not reject H_0" rather than "accept H_0" emphasizes that this single value is merely one of many plausible values, because of the inherent sampling variability.
- The terminology "accept H_a" is permissible because when the P-value is sufficiently small, all the plausible values fall within the infinite range of values that H_a specifies.

Type I and Type II Errors

Because of sampling variability, decisions in tests always have some uncertainty.

- A decision could be erroneous.
- The two potential errors are called *Type I error* and *Type II error*.
- When H_0 is true, a **Type I error** occurs when H_0 is rejected.
- When H_0 is false, a **Type II error** occurs when H_0 is not rejected.

	Reject H_0	Do not reject H_0
H_0 true	Type I error	Correct decision
H_0 false	Correct decision	Type II error

Type I Error

When we make a decision with α-level $= 0.05$, we reject H_0 when the P-value is less than or equal to 0.05.

- When H_0 is true, for any test statistic that has a continuous distribution, the P-value has a uniform distribution over the interval $[0, 1]$. Therefore, when H_0 is true, the probability that the P-value is less than or equal to 0.05 is and we thus incorrectly reject H_0 is 0.05.
- We can control for P(Type I error) by the choice of α.
- The more serious the consequences of a Type I error, the smaller α should be.
- In practice, $\alpha = 0.05$ is common, just as a confidence level of 0.95 is common with confidence intervals. When a Type I error has serious implications, it is safer to use $\alpha = 0.01$.
- When we make a decision, we do not know whether we have made an error, just as we do not know whether a particular confidence interval truly contains the parameter value. However, we can control the probability of an incorrect decision for either type of inference.

Type I Error and Type II Error

The collection of test statistic values for which a significance test rejects H_0 is called the **rejection region**.

- For example, for two-sided tests about a proportion p with $\alpha = 0.05$, the P-value is less than or equal to 0.05 whenever the test statistic $|z| \geq 1.96$.
- For a one-sided test with $H_a : p > p_0$, the rejection region is $z \geq z_{0.05} = 1.645$.
- The cutoff point for the rejection region is called the **critical value**.
- With $\alpha = 0.05$, the probability of the values in the rejection region under the standard normal curve that applies for the sampling distribution of z when H_0 is true is 0.05, reflecting that the probability of Type I error (i.e., rejecting H_0 when it is true) is exactly the α-level of 0.05.

As P(Type I Error) Decreases, P(Type II Error) Increases

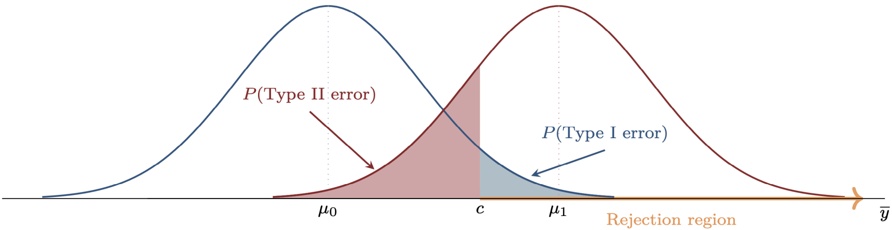
Depending on the nature of the application, one type of error may be considered more serious than the other.

- When we make a decision for which a Type I error would be especially harmful,. when not use an extremely small P(Type I error), such as $\alpha = 0.0000001$?

Consider this in the context of a criminal legal trial. Let H_0 represent defendant innocence and H_a represent guilt.

- The jury rejects H_0 and judges the defendant to be guilty if the evidence is sufficient to convict. A Type I error, rejecting a true H_0, occurs in convicting a defendant who is actually innocent.
- We don't make it nearly impossible to convict someone who is innocent, because we would then be unlikely to convict many defendants who are truly guilty.
- In other words, when P(Type I Error) decreases, P(Type II Error) increases

As P(Type I Error) Decreases, P(Type II Error) Increases



The left curve is the sampling distribution of $\overline{Y}$ when $H_0 : \mu = \mu_0$ is true. The right curve is the sampling distribution when a particular value μ_1 from H_a is the actual value of μ.

The rejection region consists of $\overline{y}$ values above some critical value c, where c depends on our selection for $\alpha = P$(Type I error), which is the probability under the left curve to the right of c (i.e., the blue-shaded area in the figure).

The P(Type II error) is the probability to the left of the critical value c under the right curve (i.e., the red-shaded area in the figure). When we make α smaller, we move the critical value c to the right, but then P(Type II error) increases.

As P(Type I Error) Decreases, P(Type II Error) Increases

For a fixed P(Type I error), P(Type II error) has more than one value, because H_a contains a range of possible values for μ_1.

- As μ_1 moves further away from μ_0 in the figure, the right curve moves to the right, and the probability under that curve to the left of the critical value c decreases.
- The farther that μ_1 falls from μ_0, the less likely a Type II error.
- For a fixed P(Type I error), P(Type II error) also depends on the sample size. As n increases, the curves in the figure are narrower, and a Type II error is less likely.
- Keeping both P(Type I error) and P(Type II error) at low levels requires using a relatively large n unless the parameter falls quite far from the H_0 value.

Type II Error

In summary, P(Type II error) decreases as

- P(Type I error) increases
- The true parameter value is farther from the value in H_0
- The sample size n increases

When H_0 is false, the probability of (correctly) rejecting H_0 is called the **power** of the test.

- For a particular value of the parameter from within the H_a range,

$$\text{Power} = 1 - P(\text{Type II error}).$$

Example: Astrology

One scientific test of the pseudo-science astrology used the following experiment:

- For each of 116 adult subjects, an astrologer prepared a horoscope based on the positions of the planets and the moon at the moment of the person's birth.
- Each subject also filled out a California Personality Index survey.
- For each adult, their birth data and horoscope were shown to an astrologer with the results of the personality survey for that adult and for two other adults randomly selected from the experimental group.
- The astrologer was asked which personality chart of the three subjects was the correct one for that adult, based on their horoscope.

Example: Astrology

Let p denote the probability of a correct prediction. If the predictions are no different from random guessing, then $p = 1/3$.

- To test this against the alternative that the predictions are better than random guessing, we test $H_0 : p = 1/3$ against $H_a : p > 1/3$.
- The National Council for Geocosmic Research, which supplied the astrologers for the experiment, claimed p would be 0.50 or higher.

We would like to find the probability of a Type II error (failing to reject H_0 when it is false) if $p = 0.50$ for an $\alpha = 0.05$-level test.

Example: Astrology

We first find the sample proportion $\overline{y}$ values for which we would not reject H_0.

- Under $H_0 : p = 1/3$ and with $n = 116$, the sampling distribution of $\overline{Y}$ has standard error

$$se_0 = \sqrt{\frac{\frac{1}{3}(1 - \frac{1}{3})}{116}} = 0.0438.$$

- The P-value (right-tail probability) equals 0.05 if the test statistic $z = 1.645$, which is the critical value.

- On the sample-proportion scale, the critical value is

$$1/3 + 1.645 * se_0 = 0.405.$$

Example: Astrology

To find $P(\text{Type II error})$, if $p = 0.50$, we need to find $P(\overline{Y} < 0.405)$ if the sampling distribution of $\overline{Y}$ is based on $p = 0.50$.

- If $p = 0.50$, the standard error of $\overline{Y}$ is

$$\sqrt{\frac{\frac{1}{2}(1-\frac{1}{2})}{116}} = 0.0464.$$

- Then $P(\overline{Y} < 0.405)$ is the probability that a standard normal random variable is less than

$$\frac{0.405 - 0.50}{0.0464} = -2.04.$$

- The probability that a standard normal random variable falls below -2.04 is 0.02.
- If the astrologers truly had the predictive power they claimed ($p = 0.50$), the chance of failing to detect this with this experiment would have only been about 0.02.

Example: Astrology

Recall that as $\alpha = P(\text{Type I error})$ decreases, $P(\text{Type II error})$ increases.

- If the astrology study used $\alpha = 0.01$, when $p = 0.50$ you can verify that $P(\text{Type II error}) = 0.08$, compared to $P(\text{Type II error}) = 0.02$ when $\alpha = 0.05$.
- Recall also that $P(\text{Type II Error})$ increases when p is closer to H_0.
- For example, when p decreases from 0.50 to 0.40, with $\alpha = 0.05$, $P(\text{Type II Error})$ increases from 0.02 to 0.55.
- In general $P(\text{Type II error})$ can be found with software or approximated by simulation.

Duality between Significance Tests and Confidence Intervals

Conclusions using two-sided significance tests are consistent with conclusions using confidence intervals.

For example, the significance test of $H_0 : \mu = \mu_0$ versus $H_a : \mu \neq \mu_0$ rejects H_0 at the $\alpha = 0.05$ level when the observed value of the test statistic $t = (\overline{y} - \mu_0)/se$ is greater in absolute value than $t_{0.025}$, which is when $\overline{y}$ falls more than $t_{0.025} * se$ from μ_0.

- But if this happens then the 95% confidence interval for μ, which is $\overline{y} \pm t_{0.025} * se$, does not contain μ_0.

Effect of Sample Size: Statistical vs. Practical Significance

In a previous example, we considered a 7-point scale of political ideology, with level 4 for "moderate."

- We saw that for a sample of 369 Hispanic Americans, $\overline{y} = 4.089$.
- We can also analyze the entire sample, combining race/ethnicity categories, to illustrate the effect of sample size on results of a test.
- The $n = 2575$ observations have $\overline{y} = 4.108$ and $s = 1.425$.
- We test $H_0 : \mu = 4.0$ against $H_a : \mu \neq 4.0$ using a test statistic with observed value

$$t = \frac{\overline{y} - 4.0}{1.425/\sqrt{2575}} = 3.85,$$

and the two-sided P-value is 0.0001, providing very strong evidence that μ falls on the "conservative" side of "moderate."

Effect of Sample Size: Statistical vs. Practical Significance

The two-sided P-value is 0.0001, providing very strong evidence that μ falls on the "conservative" side of "moderate."

- On a scale of 1.0 to 7.0, however, $\overline{y} = 4.108$ is close to 4.0.
- Although the difference of 0.108 between $\overline{y}$ and $\mu_0 = 4.0$ is highly significant statistically, it is small in practical terms.
- The difference is highly statistically significant but not practically significant.
- The 95% confidence interval for μ is (4.05, 4.16).
- By contrast, if $\overline{y}$ had been 6.108 instead of 4.108, the 95% confidence interval of (6.05, 6.16) would indicate a substantial practical difference from 4.0.

Effect of Sample Size: Statistical vs. Practical Significance

The analysis for the $n = 369$ Hispanic Americans with $\overline{y} = 4.1$ has a P-value of 0.20, not much evidence against H_0, but now with $\overline{y} = 4.1$ for $n = 2575$, the P-value is 0.0001.

- A particular difference between a parameter estimate and the value of the parameter under H_0 has a larger test statistic and a smaller P-value as n increases, because the standard error in the denominator of the test statistic decreases.
- The larger n is, the more certain we can be that sample deviations from H_0 are indicative of true population deviations.
- Even a small difference can yield a small P-value, however, if n is sufficiently large.
- For large samples, statistical significance does not imply practical significance. A tiny P-value, such as 0.0001, does not imply an important finding in any practical sense.
- The tiny P-value merely means that if H_0 were true, the observed data would be highly unusual. It does not mean that the parameter falls far from H_0 in practical terms.

Effect of Sample Size: Statistical vs. Practical Significance

To summarize practical significance, we can report an effect size measure:

- For a single sample, the effect size is defined as follows

$$\text{effect size} = \frac{\overline{y} - \mu_0}{s}.$$

- In this example, for the entire sample of $n = 2575$, the estimated effect size is $(4.108 - 4.0)/1.425 = 0.08$, which is tiny.

Significance Tests are Less Useful than Confidence Intervals

In most fields of application, the usual null hypotheses that have a single value in H_0, such as $H_0 : \mu = 0$ or $H_0 : \mu_1 - \mu_2 = 0$, are rarely true.

- That is, rarely is the parameter exactly equal to the value listed in H_0.
- With sufficiently large samples, so that a Type II error is unlikely, these hypotheses will normally be rejected.
- Then, of main relevance is the direction of the effect (e.g., whether $\mu_1 > \mu_2$ or $\mu_2 > \mu_1$) and whether the parameter is sufficiently different from the H_0 value to be of practical importance.
- Although P-values are useful for summarizing the evidence against H_0, significance tests are overemphasized in much research.

Significance Tests are Less Useful than Confidence Intervals

We learn more from confidence intervals.

- A test merely indicates whether the particular value in H_0 is plausible.
- The confidence interval, by contrast, displays the entire set of plausible values.
- It shows the extent to which reality may differ from H_0 by showing whether the values in the interval are far from H_0.
- Thus, it helps us to determine whether rejection of H_0 has practical importance.
- When a P-value is not small but the confidence interval is quite wide, this forces us to realize that the parameter might well fall far from H_0 even though we cannot reject H_0.
- This also supports why it does not make sense to "accept H_0" (rather than "fail to reject H_0").

Likelihood-Ratio Tests

We have formed confidence intervals and test statistics for significance tests using a pivotal quantity.

- There is an alternative method that applies more generally, such as to parameters of probability distributions that may not refer to means or proportions.
- It uses the likelihood function directly to formulate significance tests and corresponding confidence intervals.
- This method assumes a particular family of distributions for the data, since the likelihood function derives from the joint distribution of the observations.

Likelihood-Ratio Tests

For a generic parameter θ, the test of $H_0 : \theta = \theta_0$ uses the likelihood function $L(\theta)$ through the ratio of:

1. The maximum of $L(\theta)$ over all possible values of θ — this is $L(\hat{\theta})$, where $\hat{\theta}$ is the maximum likelihood estimate
2. The value of $L(\theta)$ when $H_0 : \theta = \theta_0$ is true — this is $L(\theta_0)$

Note that $L(\hat{\theta})$ is always at least as large as $L(\theta_0)$. This leads to the **likelihood-ratio test statistic:**

$$2 \log[L(\hat{\theta})/L(\theta_0)] = 2[l(\hat{\theta}) - l(\theta_0)],$$

where $l(\cdot)$ denotes the log-likelihood function.

This test statistic is nonnegative, and larger values of $L(\hat{\theta})/L(\theta_0)$ yield larger values of the test statistic, and stronger evidence against H_0.

Likelihood-Ratio Tests

The reason for taking the log transform of the likelihood ratio and multiplying it by two is that, under $H_0 : \theta = \theta_0$, for large samples $2 \log[L(\hat{\theta})/L(\theta_0)]$ has approximately a chi-squared distribution with $df = 1$.

- Recall that with $df = 1$, the χ_1^2 distribution has mean 1 and is the distribution of Z^2 for a standard normal random variable Z.
- Since larger values of the test statistic provide stronger evidence against H_0, the P-value is the right-tail probability from the χ_1^2 distribution of values at least as large as the observed test statistic.
- The right-tail probability for the χ_1^2 distribution above a point x equals the two-tail probability for the standard normal below $-\sqrt{x}$ and above $\sqrt{x}$. A one-sided χ_1^2 P-value is a two-sided standard normal P-value.

Likelihood-Ratio Tests

The likelihood-ratio test has a corresponding confidence interval.

- The 95% confidence interval for θ is the set of θ_0 values for $H_0 : \theta = \theta_0$ such that the P-value in the test is larger than 0.05.
- With multiple parameters, it is called a *profile likelihood confidence interval*, because it profiles out the other parameters by using their values that maximize the likelihood at any fixed value θ_0 for the main parameter of interest.

Nonparametric Tests

For comparing two means, we considered a significance test method presented that assumes that the population distributions are normal, with the same variance.

- Although the two-sided case is robust to violations of this assumption, in some cases we might prefer to compare groups without it.
- One such case is when the method is not robust, such as for one-sided tests with small samples when the population distribution is highly skewed.
- With small sample size n, it is difficult to judge the shape of a population distribution.
- Also, if that distribution may be highly skewed, the mean is less relevant as a summary measure, but we may still want to test the hypothesis that two groups have identical population distributions.

Nonparametric Tests

As an alternative, we can consider tests based on nonparametric methods, which are inferential methods that do not assume a particular family of probability distributions for the population.

For these significance tests, H_0 states that the population distributions are identical for the two groups, without specifying their shape.

The tests assume independent observations from the two groups, obtained with randomization.

A Permutation Test to Compare Two Groups

A **permutation test** derives the sampling distribution of a statistic such as the difference between the sample medians or sample means by evaluating all possible permutations for how the $n_1 + n_2$ observations in the two samples could have been allocated to the two groups, with n_1 observations in the first group and n_2 in the second.

When the null hypothesis H_0 of identical population distributions is true, all such permutations are equally likely.

The P-value is the proportion of them for which the chosen statistic is at least as extreme as observed, with the more extreme values based on whether the test has a one-sided or two–ided alternative.

A Permutation Test to Compare Two Groups

As n_1 and n_2 increase, it is computationally more demanding to evaluate all possible permutations, but software and apps can generate a very large random sample (say, a million) of them.

The proportion of those simulated permutations that are at least as extreme as the observed samples in terms of the chosen statistic serves as a close approximation for the exact permutation P-value.

Example

In a study to determine whether dogs prefer petting or vocal praise, the researchers randomly placed 14 dogs into two groups of 7 each.

- In group 1, the owner would pet the dog.
- In group 2, the owner would provide vocal praise.

The response variable is the time, in seconds, that the dog interacted with its owner.

- Response variable in group 1: 114, 203, 217, 254, 256, 284, 296
- Response variable in group 2: 4, 7, 24, 25, 48, 71, 294

The lowest value in group 1 and the highest value in group 2 seem to be outliers. As the true distributions are potentially highly skewed, we focus on the median, which is 254 seconds in group 1 and 25 seconds in group 2. Note that the difference between medians is $254 - 25 = 229$.

Example

```
> library(EnvStats)
> petting <- c(114, 203, 217, 254, 256, 284, 296)
> praise  <- c(4, 7, 24, 25, 48, 71, 294)
> test <- twoSamplePermutationTestLocation(petting, praise,             # compare medians
+         fcn="median", alternative="greater", exact=FALSE, n.permutations=1000000)
> test$p.value
[1] 0.01752 # simulated P-value for Ha: higher median for petting
> test <- twoSamplePermutationTestLocation(petting, praise, fcn="mean", # compare means
+         alternative="greater", exact=FALSE, n.permutations=1000000)
> test$p.value
[1] 0.0038  # simulated P-value for Ha: higher mean for petting
```

Here we have results based on one million simulations. The evidence is quite strong that the population median is higher for petting, and this is true also for the population means, as shown in the output.

It is also possible to obtain an exact P-value using the app `artofstat.com/web-apps`. This app indicates that 60 of the 3432 permutations had a sample median difference of at least 229, so the exact P-value is $60/3432 = 0.01748$.

Example

We test the null hypothesis of identical population distributions against the one-sided alternative that the population median is higher for petting.

- For the test statistic, we use the difference between the two sample medians.
- The number of ways to partition the 14 dogs into two groups of 7 each is $\binom{14}{7} = 3432$. The P-value for the permutation test is the proportion of those partitions for which the difference between the two sample medians is at least $254 - 25 = 229$.
- In `R`, a function in the `EnvStats` package can conduct this test, but it uses the exact distribution only when $n_1 + n + 2 \leq 10$ and otherwise uses simulation.

Wilcoxon Test: Comparing Mean Ranks for Two Groups

Many nonparametric statistical methods use the rankings of the observations, taking the $n_1 + n_2$ observations and ranking them in magnitude from 1 to $n_1 + n_2$.

- Observations having the same values, said to be tied, are assigned the average of the ranks that would have been given without the ties.
- An advantage of using ranks is that outlying observations are no longer outliers when converted to ranks and are not so highly influential on measures such as the mean.
- The best known rank-based significance test for comparing two groups is the **Wilcoxon test**. For it, H_0 states that the two population distributions are identical and H_a states that one distribution is a location shift of the other (can be one-sided or two-sided).
- The evidence about H_0 is summarized by the difference between the mean ranks for the two groups.

Wilcoxon Test: Comparing Mean Ranks for Two Groups

For a study using randomization, the P-value is the probability, under H_0, of a difference between the sample mean ranks like the observed difference or even more extreme, in terms of favoring H_a.

- To find the P-value, software enumerates all possible ways of partitioning the observations, n_1 in the first group and n_2 in the second, and finds the proportion of the partitionings for which the difference between the sample mean ranks is at least as extreme as observed.
- When it is not feasible to generate all the permutations, software can closely approximate the exact P-value by taking a very large random sample of them.
- For large n_1 and n_2, the sampling distribution of the difference between the sample mean ranks is approximately normal, and some software bases the P-value on tail probabilities for this normal approximation.

Example

For the study about dog petting and praise, the times that the dog interacted with the owner are $(114, 203, 217, 254, 256, 284, 296)$ seconds in group 1 (petting) and $(4, 7, 24, 25, 48, 71, 294)$ seconds in group 2 (praise).

- We use the one-sided H_a of a higher distribution for petting than for vocal praise.
- Starting the ranks with the lowest times, the ranks for group 1 are (7, 8, 9, 10, 11, 12, 14), for a sum of ranks of 71. The smallest value it can possibly take is $n_1(n_1 + 1)/2 = 7(8)/2 = 28$.
- A higher sum of ranks for the petting group, and equivalently a greater difference between sample mean ranks for petting and for praise, give more evidence in favor of petting.

Example

The R code for doing this analysis is:

```
> petting <- c(114, 203, 217, 254, 256, 284, 296)
> praise  <- c(4, 7, 24, 25, 48, 71, 294)
> wilcox.test(petting, praise, alternative="greater", exact=TRUE)
```

with output:

```
W = 43, p-value = 0.008741 # exact P-value for Ha favoring petting
alternative hypothesis: true location shift is greater than 0
```

The W statistic reported is the sum of ranks for group 1 minus its smallest possible value, $W = 71 - 28 = 43$. The P-value gives strong evidence in favor of a higher distribution of times for petting.

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.